Comparison Operators in WHERE

Lecture Notes · Game DB Junior

In this lesson

- The **comparison operators** we can use inside WHERE
- How to check if a value is **equal** (=) to a constant value
- How to check if a value is **not equal** (!= or <>)

1. A quick reminder

The **WHERE** keyword lets us filter the results we get back from a query. In the last lesson we checked the area of each row, and we only returned the row if its area was greater than 4000:

```
SELECT name, area FROM cities WHERE area > 4000;
```

2. Comparison operators

Besides the greater than sign (>), there are many other **comparison operators** we can use inside a WHERE statement. We can check if two values are equal, if one value is less than another, if values are not equal, and even if a value is between other values or inside a list.

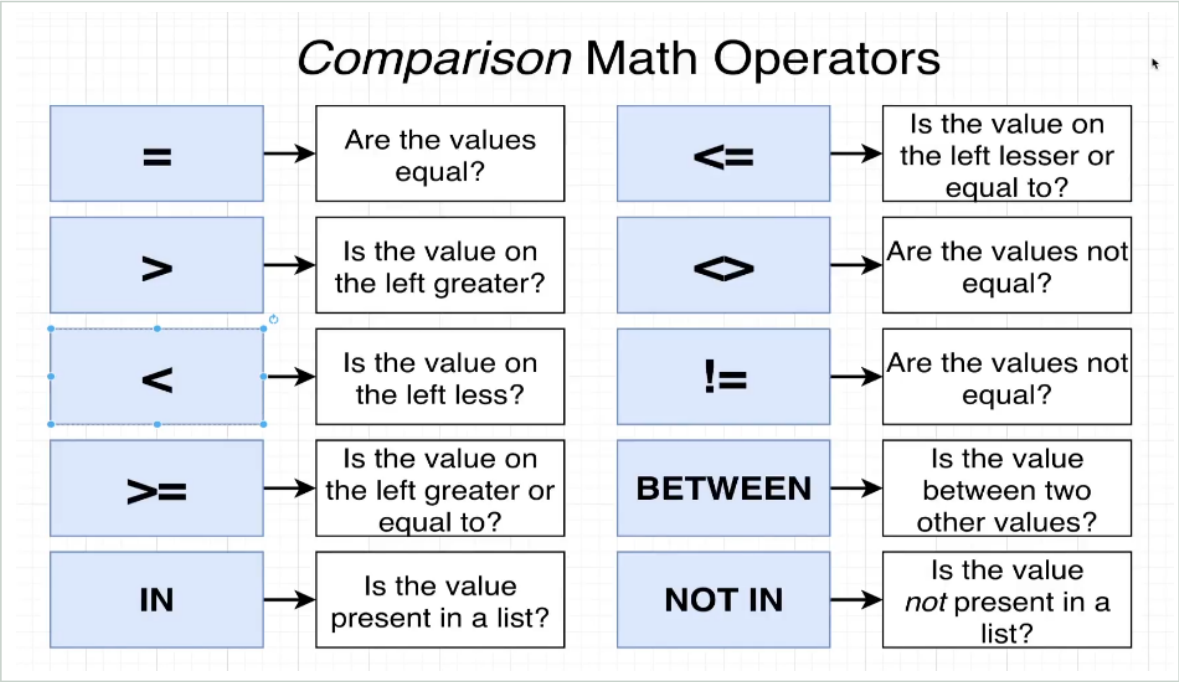


Figure 1 — Comparison math operators

Operator	Question it asks
=	Are the values equal?
>	Is the value on the left greater?

Operator	Question it asks
<	Is the value on the left less?
>=	Is the value on the left greater or equal to?
<=	Is the value on the left lesser or equal to?
<>	Are the values not equal?
!=	Are the values not equal?
BETWEEN	Is the value between two other values?
IN	Is the value present in a list?
NOT IN	Is the value *not* present in a list?

3. Equal (=)

We check if a column value is equal to a **constant value**:

```
SELECT name, area FROM cities WHERE area = 8223;
```

Note: In many other programming languages, a single equal sign means **assignment**. Here we are **not** assigning anything. We are doing a **comparison**: we only want to see the rows where the area is equal to 8223.

Tokyo is the only row in the cities table with an area of 8223. So when we run it, we get only **Tokyo**.

4. Not equal (!= and <>)

We can also find all the rows that do **not** have an area of 8223. There are two ways to write it:

```
SELECT name, area FROM cities WHERE area != 8223;

SELECT name, area FROM cities WHERE area <> 8223;
```

- **!=** is an exclamation mark and an equal sign.
- **<>** is a less than sign and a greater than sign. It does the **same thing**, it is only written in a different way.
- Both give us every city that does not have an area of 8223. The result is **exactly the same**.

Keep in mind: the comparison runs for **every single row** in the table.

5. Running the code (my practice)

In my table, Tokyo has an area of **8231**, so I used 8231 in my queries.

Query 1: area equal to 8231.

```sql
-- Show the name and area of cities with an area equal to 8231
SELECT
    name, area
FROM cities
WHERE area = 8231;
```

Figure 2 — WHERE area = 8231

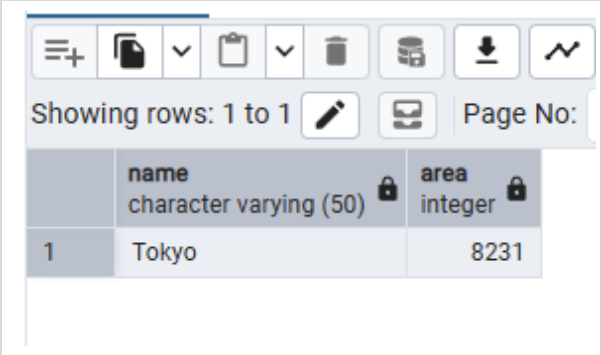


Figure 3 — Result: only Tokyo, 1 row

Query 2: area not equal to 8231, written two ways (!= and <>).

```sql
-- Show the name and area of cities with an area not equal to 8231
SELECT
    name, area
FROM cities
WHERE area != 8231;



-- Show the name and area of cities with an area not equal to 8231
SELECT
    name, area
FROM cities
WHERE area <> 8231;
```

Figure 4 — WHERE area != 8231 and WHERE area <> 8231

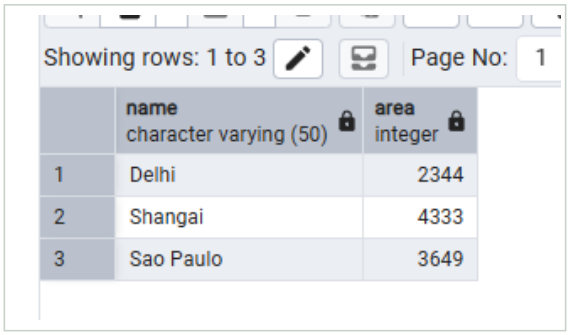


Figure 5 — Result: Delhi, Shangai, Sao Paulo, 3 rows

6. Summary

Idea	Remember
Comparison operators	= > < >= <= <> != BETWEEN IN NOT IN
=	A comparison, not an assignment. Are the values equal?
!= and <>	Both mean not equal. Same result
Every row	The comparison runs for every single row in the table

Next lesson: more advanced checks with BETWEEN, IN and NOT IN.

Sources
Lecture transcript and lecture image (Figure 1)
My own practice screenshots (Figures 2–5)